SOHAM VYAS

AI Systems Engineer & Open Source Contributor (SymPy / GSoC 2027)

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EDUCATION

Indian Institute of Technology Jodhpur (IIT Jodhpur) — *BS in Data Science & Artificial Intelligence*

Relevant Coursework: Distributed Systems, Computer Algebra, Linear Algebra, Machine Learning, Data Structures & Algorithms.

OPEN SOURCE CONTRIBUTIONS

SymPy (Computer Algebra System in Python) — *Contributor & GSoC 2027 Candidate*

- **KroneckerProduct.trace() Optimization:** Formulated and upstreamed exact tensor trace evaluation algorithm. Eliminated naive O(n^3) matrix expansion into O(1) symbolic scalar property tr(A ⊗ B) = tr(A) * tr(B), saving 99.999% FLOPs and eliminating OutOfMemory crashes on large matrices.
- **MatrixUnit.trace() Symbolic Handler:** Implemented dispatch logic for elementary matrix units with non-square and evaluate=False guardrails, passing 150+ SymPy matrix test suites.

FEATURED ENGINEERING PROJECTS

Cybersecurity Risk Analyzer (Python, AST, Shannon Information Theory) — *GitHub: 22+ Stars*

- Built an automated source code static auditing tool utilizing Claude Shannon's mathematical entropy equation E = L * log2(R) to detect hardcoded credentials and high-entropy API secrets.
- Integrated NIST 800-63B standards and 100 GH/s GPU cluster brute-force duration modeling directly into CLI and browser environments.

AI Rate Limiter Shield (FastAPI, Python, Redis, Atomic Lua)

- Engineered an atomic sliding-window rate limiter using Redis Lua scripting to prevent race conditions during high concurrency (10,000+ req/sec).
- Eliminated boundary double-burst vulnerabilities and token exhaustion attacks with sub-millisecond execution latency (0.38 ms).

SchemaGuard (Universal Database Schema Migration & Validation)

- Developed a multi-database schema synchronization engine supporting SQLite, PostgreSQL, and MySQL with transactional rollback guarantees.

Neural TTS & Indian Accent Speech Pipeline (PyTorch, Deep Learning)

- Built an offline neural voice engine calibrated for Indian regional phonetic accents with lightweight on-device inference.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, Lua, SQL, Bash, R
Backend & Systems: FastAPI, Flask, Redis (Atomic Lua), Docker, Linux, Node.js, Git, CI/CD
Math & AI: Computer Algebra (SymPy), PyTorch, Machine Learning, Information Theory, Linear Algebra
Web Technologies: React, Next.js, Tailwind CSS, Turbopack, REST APIs, WebSockets